

## PS 26132 — Team Explanation Points

### 1. What is the actual problem?

- Farmers, especially small farmers/FPOs, don't have a clear picture of **where they should sell**.
- They may know the mandi price, but not:
  - which buyer is better
  - transport cost
  - quality requirements
  - payment reliability
  - storage opportunity
  - whether to sell now or wait
- Farmers may therefore make **distress sales immediately after harvest**.
- Buyers also struggle to find **consistent quantity + required quality**.

### 2. Important insight

**The problem is not simply lack of price information.**

The real problem is:

**“Given my crop, quantity, quality, location and constraints, what is the best way to sell?”**

---

### 3. We checked existing systems

#### e-NAM

Already provides:

- digital agricultural trading
- price discovery
- quality/assaying
- FPO aggregation
- logistics
- online payments
- advance demand/supply
- warehouse-based trading

**Conclusion:** We cannot simply make another e-NAM.

**AgriBazaar**

Already provides:

- farmer-buyer marketplace
- buyer/seller verification
- negotiation
- quality
- logistics
- payments
- auctions
- market intelligence

**Conclusion:** A normal agricultural marketplace would be too similar.

### **Maharashtra/MSAMB**

Already provides:

- mandi prices
- farmer/FPC/buyer ecosystem
- farmer/FPC inventory
- buyer enquiries
- grading/certification information
- market information

**Conclusion:** Even a Maharashtra-specific marketplace is not sufficiently novel.

---

## **4. So what is the gap?**

Existing systems mainly help with:

**“What is the market situation?”**

or

**“Where can I trade?”**

Our proposed system should answer:

**“What should I do?”**

---

## **5. Our proposed idea**

### **Agricultural Market Decision Intelligence System**

Farmer/FPO enters:

- crop
- quantity
- location
- quality
- harvest date
- cash requirement
- storage availability

System combines:

- mandi prices
- historical prices
- market arrivals
- buyer demand
- logistics cost
- storage options
- quality requirements
- price forecast
- buyer reliability

Then recommends:

**Best selling strategy**

For example:

**Sell to Buyer A now**

OR

**Store for 7 days**

OR

**Aggregate with another FPO**

OR

**Sell through another market**

OR

**Split the lot between multiple buyers**

---

**6. Example to explain to team**

Suppose an FPO has:

**100 tonnes soybean**

Three options:

| Option       | Gross Price       | Costs/Risk       | Expected Net |
|--------------|-------------------|------------------|--------------|
| Local Mandi  | ₹4,900/q          | Low              | ₹4,780       |
| Buyer A      | ₹5,150/q          | Transport + risk | ₹4,930       |
| Store 7 days | ₹5,300/q expected | Storage + risk   | ₹4,980       |

System doesn't simply say:

“Buyer A has highest price.”

It calculates the **actual expected realization** and recommends the best option.

---

## 7. Our strongest feature

### Sell vs Store vs Aggregate

The system evaluates:

#### SELL NOW

vs

#### WAIT/STORE

vs

#### AGGREGATE

vs

#### ALTERNATIVE MARKET/BUYER

based on the FPO/farmer's constraints.

---

## 8. Second strong feature — Buyer intelligence

Instead of ranking buyers only by price:

Buyer A

₹5,300

Payment reliability: 70%

Distance: 200 km

Buyer B

₹5,150

Payment reliability: 97%

Distance: 70 km

System may recommend **Buyer B** because the overall expected outcome is better.

So:

**Highest price ≠ best buyer.**

---

### **9. Third strong feature — FPO aggregation**

Example:

Buyer requires: 100 tonnes

FPO A → 30 tonnes

FPO B → 40 tonnes

FPO C → 35 tonnes

System identifies:

**105 tonnes can satisfy the buyer's requirement.**

Then helps create a collective lot.

This improves:

- bargaining power
  - transportation efficiency
  - buyer access
  - volume consistency
- 

### **10. Fourth — Market stress prediction**

Government dashboard can identify:

**Expected supply ↑ + demand ↓ + storage shortage → distress-sale risk**

For example:

“Nashik onion market may face supply pressure next week.”

Government/FPOs can then look for:

- alternative buyers

- storage
- alternate markets
- logistics
- aggregation

This makes the project useful not only to farmers but also to the **Maharashtra government**.

---

## 11. What data can we actually get?

This was an important part of our research.

### ● Available

Government sources can provide:

- mandi prices
- minimum/maximum/modal prices
- commodity
- variety
- grade
- market
- dates
- market arrivals
- historical data
- crop production data

Main sources:

**AGMARKNET / data.gov.in / e-NAM / Maharashtra market systems**

### ● We generate ourselves

- buyer demand
- FPO supply
- storage availability
- farmer urgency
- quality information
- offers

### ● Future integrations

- e-NAM

- MSAMB
  - WDRA
  - IMD
  - logistics providers
- 

## 12. Very important: what we DON'T need to do

Tell the team:

- ✗ Don't build another marketplace.
  - ✗ Don't build just a mandi-price app.
  - ✗ Don't build only an AI price predictor.
  - ✗ Don't add blockchain just for "innovation."
  - ✗ Don't claim smartphone image = certified quality testing.
- 

## 13. Where AI/ML actually fits

We can use ML for:

### Price forecasting

What could the price range be in the next few days?

### Arrival forecasting

How much supply may enter a market?

### Opportunity ranking

Which selling option is likely to give the best outcome?

But the final recommendation should combine:

### ML + optimization + real-world constraints

rather than blindly using deep learning everywhere.

---

## 14. Why I think this PS is worth keeping

### Strengths

- Government-backed problem
- Software-only
- Real societal impact
- Real government data available

- Strong AI/ML potential
- Optimization problem
- FPO angle
- Scalable beyond one crop
- Good SIH demo potential

#### **Main risk**

The agriculture-market ecosystem is already crowded.

Therefore:

**Our differentiation must be the decision engine, not the marketplace.**

---

#### **15. One-line explanation to your team**

If you only get **20 seconds**, say this:

**“e-NAM, AgriBazaar and Maharashtra systems already provide prices, marketplaces and trading infrastructure. We shouldn't build another marketplace. Our idea is an intelligence layer that takes a farmer/FPO's crop, quantity, quality, location, liquidity and storage constraints, compares available market/buyer options, and recommends whether to sell now, wait, store, aggregate or switch buyers/markets for the best risk-adjusted realization.”**

#### **Current assessment**

**PS 26132 → 8.5/10 data feasibility → SHORTLIST**